Retention-constrained class expansion with reference-anchored candidate tables

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Abstract

Expandable recognition must admit new categories while preserving an explicit accuracy requirement on the original classes. We study novel recognition at a common original-class constraint and the predictive value of retaining a classifier across successive arrivals. Reference-anchored candidate tables combine trained base columns, semantic anchors and novel visual prototypes in a persistent score frame. Base-only pseudo-expansion episodes calibrate the mixtures; validation-selected admission thresholds are evaluated separately from descriptive matched-test frontiers. Experiments cover CIFAR-100, Oxford-IIIT Pets and 200-class ImageNet-R, ten class partitions per dataset and two frozen CLIP architectures. On ImageNet-R with ViT-L/14, a common base accuracy of 86.42% permits 84.31% novel accuracy. Across five arrivals, retaining the classifier adds 0.71 percentage points over dense prototype fusion and 1.42 points over class-mean semantic-cache retrieval in average incremental accuracy. The prototype comparison transfers to ViT-B/32 with a 0.90-point gain. Paired confidence intervals characterize class-partition variation conditional on the fixed data and encoders. Raw-score and fixed-temperature controls isolate reference normalization, while complete validation-policy results distinguish empirical feasibility from attained test retention. The evidence identifies the operating regimes in which preserving trained discrimination supplies additional class-expansion capacity.

Keywords: Class expansion, Retention constraints, Incremental learning, Evidence fusion, Vision–language models, Long-tailed recognition

1. Introduction

Recognition systems often acquire new categories after deployment, while the original classifier has been fitted to imbalanced supervision. Long-tailed recognition addresses the

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unequal evidence available to common and rare classes [1, 2, 3]. Class expansion must put that learned discrimination to work alongside the names and sparse examples supplied for arriving categories.

Pre-trained vision–language models provide a shared interface for this evidence. CLIP maps images and descriptions into an embedding space [4]; prompt adaptation and lightweight fine-tuning specialize its predictions [5, 6, 7]; caches and prototypes incorporate labeled support [8, 9, 10]. An expanded classifier can therefore retain trained base columns and append semantic anchors and prototypes for novel classes.

The practical criterion is novel recognition under a stated base-accuracy requirement. Fixing this requirement identifies how much new-class recognition the system can admit while preserving its established service level. Successive class arrivals introduce a second criterion: the paired gain obtained by retaining the base-trained classifier throughout the sequence. Separating a fixed-gate removal from a recalibrated classifier-free alternative measures both the contribution of the stored learned evidence and its value after calibration is restored.

We address these criteria with *reference-anchored candidate tables*. Each class stores the evidence available to it, and each branch normalizes its scores against fixed base-class references. The table retains the original learned columns and appends new rows without refitting them. This reference frame makes the existing scores independent of vocabulary size, allowing the novel admission threshold to be analyzed separately from within-group discrimination.

Class-incremental learning studies retention of earlier knowledge [11, 12], and pre-trained prototypes and analytic readouts enable efficient class addition [13, 14]. Recent methods combine multiple feature spaces [15], calibrate prototypes through sharpness-aware learning [16], and fuse CLIP prompts across modalities [17]. Our evaluation asks what a retained classifier contributes under a common base constraint and across repeated arrivals. A score-preservation proposition supplies the mechanism; matched-retention frontiers and paired sequential comparisons supply the recognition evidence.

On the 200-class ImageNet-R task, the retained classifier adds 0.71 [0.46, 0.92] percentage points in average incremental accuracy over dense prototype fusion with ViT-L/14 and 0.90 [0.69, 1.11] with ViT-B/32. Brackets are paired 95% bootstrap intervals across ten class partitions, conditional on the fixed image pools and encoders. The expanded comparison separates classifier removal, recalibration, prototype replacement, semantic-cache retrieval and simpler score fusion. Validation-selected thresholds and their achieved test outcomes

receive a separate main-paper comparison.

The contribution is a retention-constrained fusion and evaluation framework: it organizes base-discriminative and arriving evidence in a persistent score frame, separates within-base discrimination from the admission threshold, and measures the retained classifier's paired value against recalibration and replacement. Raw and fixed-temperature controls isolate the role of the per-query reference statistics; expanded class-partition and encoder comparisons identify the operating regimes in which retaining trained evidence is useful.

2. Related work

2.1. Imbalanced supervision and heterogeneous evidence

Long-tailed recognition motivates balanced fitting when class evidence is unequal [2, 3]. CLIP supplies semantic anchors [4], prompt learning adapts textual evidence [5, 6], and prototypes and caches incorporate labeled support [9, 10, 8]. Our table combines these evidence types according to their availability, with a shared encoder and novel support budget across compared readouts.

2.2. Few-shot expansion and retained discrimination

Multi-feature space similarity supplement (MFS3) combines base and incremental feature spaces to refine boundary decisions [15]. Sharpness-aware gradient guidance (SAGG) couples flat-region optimization with prototype calibration [16]. Adaptive exploration uses reward-guided updates to balance knowledge retention and acquisition [18]. These approaches develop representations or classifiers for successive tasks. Our formulation evaluates the recognition capacity supplied by an existing classifier at a shared base-accuracy requirement. Fixed-gate removal, independent recalibration and dense prototype replacement identify distinct sources of its measured value.

2.3. Incremental recognition with pre-trained models

Class-incremental learning evaluates a model on the union of classes observed across successive training stages. Exemplar rehearsal and representation constraints are established tools for retaining earlier knowledge [11]. Few-shot class-incremental learning adds a data scarcity condition to the arriving classes [12]. Recent work studies temporal consistency with unlabeled memory replay [19] and the geometry of inter-task and intra-task discrimination [20]. These approaches motivate explicit measurement of existing-class behavior.

SimpleCIL examines prototype classifiers built from pre-trained representations [13]; RanPAC combines random projections with accumulated class statistics [14]. Reference-anchored fusion extends this setting to semantic, trained, and support-derived evidence with different availability patterns. Calibrated seen/unseen recognition [21] provides a related perspective on the base–novel operating trade-off.

Bimodal Prompt Collaborative CLIP combines attribute-enriched text with gated visual–textual prompts for few-shot incremental learning [17]. CLIP-guided continual novel class discovery uses semi-supervised prototype adaptation to transfer visual–language knowledge to a discovery model [22]. The latter operates with unlabeled novel data; our arriving classes have names and labeled support. REAL enhances representations for exemplar-free analytic learning through complementary base pretraining and feature fusion [23]. These studies establish complementary ways to retain and combine prior knowledge. Reference-anchored tables organize semantic anchors, learned columns and support prototypes within one fixed score frame, making the retained classifier's admission value directly measurable.

FeNeC represents classes with clustered feature representatives and class-specific distances, extending nearest-class-mean recognition to local neighborhoods [24]. Its Mahalanobis and Euclidean variants provide recent frozen-feature comparisons under the same base fitting data and five-shot novel support. We report native predictions and an explicitly group-offset-calibrated form.

BiMC [25] calibrates textual classifiers with fine-grained descriptions, adjusts novel visual prototypes using base prototypes, and combines modality and covariance-based metrics. HyCal [26] combines cosine and Mahalanobis information to address heterogeneous cross-discipline few-shot streams. These methods improve the evidence or metric used for incremental recognition. Our framework instead measures the contribution of a separately fitted base-discriminative branch under a shared retention requirement and preserves an explicit score frame while heterogeneous evidence is appended. The relevant comparison is the attainable base–novel trade-off after equalizing data access and calibration, rather than an unrestricted accuracy comparison across different protocols.

TV100 studies categories unavailable during CLIP's original pre-training period [27]. Here, novelty is defined relative to task-specific classifier fitting and calibration; CLIP retains its web-scale pre-training exposure.

3. Expandable recognition with heterogeneous evidence

3.1. Problem definition

Let $\mathcal{B}$ denote base classes with a labeled training set $\mathcal{D}_B$, and let $\mathcal{N}$ denote novel classes supplied at expansion time. The active vocabulary is initially $\mathcal{C}_0 = \mathcal{B}$ and subsequently becomes $\mathcal{C} = \mathcal{B} \cup \mathcal{N}$. An image encoder f maps an input image to a normalized vector $v = f(x) \in \mathbb{R}^D$. A text encoder supplies a normalized class anchor t_c. For each $c \in \mathcal{N}$, the support set $\mathcal{S}_c$ contains k labeled examples. Inference is over all active classes, without an oracle indicating whether an image belongs to a base or novel class.

Base training can be imbalanced. If n_c is the number of training images for class c, the classifier may use inverse-frequency weights or another balanced fitting procedure. After this stage, the encoder, base classifier, reference banks, and branch mixing coefficients are fixed. The group admission offset is either frozen or calibrated on base validation to enforce a retention target. At arrival, novel class information is restricted to its name and $\mathcal{S}_c$. Neither previous training images nor additional labeled novel validation images are required to construct its row.

Each row has an availability mask $m_{ec} \in \{0, 1\}$ over evidence sources e. We use semantic evidence T, learned discriminative evidence L, and visual prototype evidence P. The experimental configuration is

$$(m_{Tc}, m_{Lc}, m_{Pc}) = \begin{cases} (1, 1, 0), & c \in \mathcal{B}, \\ (1, 0, 1), & c \in \mathcal{N}. \end{cases} \tag{1}$$

Base prototypes form the visual branch's normalization references. The availability mask determines which evidence contributes to a candidate; the reference bank determines its scale.

3.2. Three notions of preservation

Preserving a parameter vector θ means that θ receives no update during expansion. Preserving a score additionally requires

$$s_{\mathcal{C}_0}(x, c) = s_{\mathcal{C}}(x, c), \quad c \in \mathcal{C}_0. \tag{2}$$

Preserving a prediction requires the maximizing class over the larger vocabulary to remain the same. The third property is stronger than the second: a newly admitted class can have a larger score than the previous winner. We seek score preservation while allowing that competition.

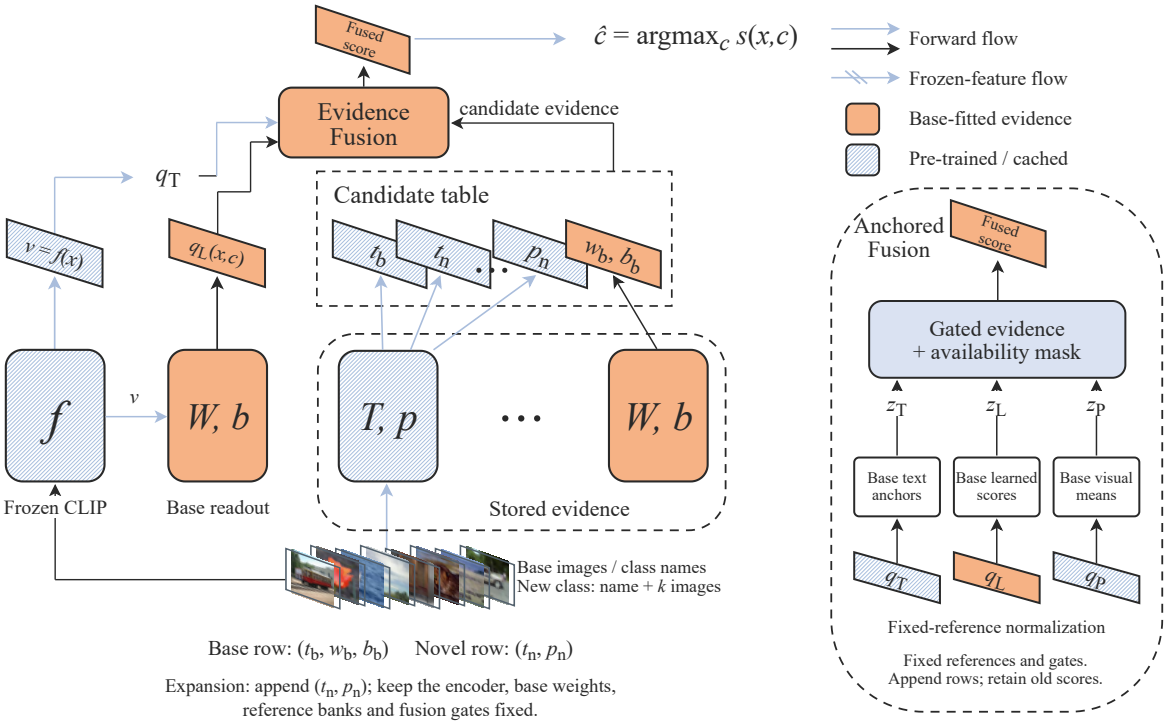


Figure 1: Reference-anchored candidate-table architecture. A shared frozen CLIP feature supplies semantic similarity and the base-trained readout. The candidate table stores text anchors for all classes, learned weights for base classes, and support prototypes for arriving classes. The inset shows the three evidence branches normalized against their respective fixed base reference sets before availability-masked fusion; $z_e = \widehat{q}_e$ denotes normalized evidence. A novel row is appended without updating the encoder, existing weights, references, or gates. Orange identifies base-fitted evidence and fusion; blue hatching identifies pre-trained or cached evidence. Branch gates remain fixed; retention control can adjust the separate novel admission offset.

The evaluation separates changes among existing classes from selection of an appended class. Under fixed evidence, reference anchoring preserves existing-class competition and admits new alternatives through their own scores.

4. Reference-anchored candidate tables

4.1. Evidence construction

The semantic score is cosine similarity between a normalized image feature and a class text anchor,

$$q_T(x, c) = v^\top t_c. \tag{3}$$

We average normalized text embeddings of eight fixed prompts and normalize the result. The same prompts construct anchors for base and arriving classes.

The learned score is the output of a base classifier,

$$q_L(x, c) = v^\top w_c + b_c, \quad c \in \mathcal{B}. \tag{4}$$

For the frozen-encoder experiments, the weights are fitted using class-balanced ridge regression. With augmented image features $\widetilde{v} = [v; 1]$, one-hot targets Y, and diagonal sample-weight matrix Ω, the solution is

$$W = (X^\top \Omega X + \lambda R)^{-1} X^\top \Omega Y, \tag{5}$$

where $R = \mathrm{diag}(1, \ldots, 1, 0)$ leaves the intercept unregularized. A sample from class c has weight $1/n_c$, giving each base class equal total fitting weight. We solve the corresponding linear system.

The visual prototype of a novel class is

$$p_c = \frac{\sum_{x \in \mathcal{S}_c} f(x)}{\| \sum_{x \in \mathcal{S}_c} f(x) \|_2}, \qquad q_P(x, c) = v^\top p_c. \tag{6}$$

The same averaging rule applied to base training features produces a reference bank $\{p_b : b \in \mathcal{B}\}$. The novel prototype stores the information acquired at class arrival; the base bank fixes the coordinate system in which its score is interpreted.

4.2. Why a changing normalization population matters

Candidate-wise standardization computes the mean and standard deviation over the current valid candidates of a branch. For text evidence, this gives

$$z_T^{\mathcal{C}}(x, c) = \frac{q_T(x, c) - \mu_T(x; \mathcal{C})}{\max(\sigma_T(x; \mathcal{C}), \epsilon)}. \tag{7}$$

Adding candidates changes μ_T and σ_T. For a text-only classifier, a common positive affine transformation preserves the ordering of the old classes. With heterogeneous fusion, however, the learned branch and text branch are rescaled differently. The relative strength of their votes can change.

A two-class example illustrates the effect. Suppose the old text scores are $(0, 2)$, giving standardized scores $(-1, 1)$, while the learned branch contributes $(0.25, -0.25)$. The second old class has the larger fused score. Appending a text score of 100 changes the two old standardized scores to approximately $(-0.728, -0.685)$. Their text separation is now about 0.043, and the unchanged learned contribution reverses the old-class ordering. No classifier weight has changed. The example isolates a normalization effect independently of whether the new class ultimately wins the expanded decision.

4.3. Fixed references for each evidence source

Let $\mathcal{R}_e$ be a fixed bank of reference rows for source e. For every query, compute

$$\mu_e(x) = \frac{1}{|\mathcal{R}_e|} \sum_{r \in \mathcal{R}_e} q_e(x, r), \tag{8}$$

$$\sigma_e^2(x) = \frac{1}{|\mathcal{R}_e|} \sum_{r \in \mathcal{R}_e} (q_e(x, r) - \mu_e(x))^2. \tag{9}$$

The normalized candidate score is

$$\widehat{q}_e(x, c) = \frac{q_e(x, c) - \mu_e(x)}{\max(\sigma_e(x), \epsilon)}. \tag{10}$$

The reference statistics remain query-dependent, which adapts the scale to the current image, but are independent of which novel candidates are active. We use base text anchors for $\mathcal{R}_T$, base classifier columns for $\mathcal{R}_L$, and base visual means for $\mathcal{R}_P$. Each bank contains 40 reference classes on CIFAR-100 and 15 on Pets. A zero-variance reference bank uses the same fixed denominator floor $\epsilon = 10^{-6}$.

The candidate score combines available normalized evidence:

$$s(x, c) = \sum_{e \in \{T, L, P\}} m_{ec} \alpha_e \widehat{q}_e(x, c) + \delta_c. \tag{11}$$

We set $\alpha_T = 1$, write $a = \alpha_L$ and $u = \alpha_P$, and use $\delta_c = d\,\mathbf{1}[c \in \mathcal{N}]$. The resulting implementation is

$$s(x, b) = \widehat{q}_T(x, b) + a\widehat{q}_L(x, b), \quad b \in \mathcal{B}, \tag{12}$$

$$s(x, n) = \widehat{q}_T(x, n) + u\widehat{q}_P(x, n) + d, \quad n \in \mathcal{N}. \tag{13}$$

The offset d calibrates competition between base and novel classes. The availability mask excludes absent learned evidence from the novel score and from reference statistics.

4.4. Score preservation under append-only expansion

Proposition 1. *Fix an input x, the evidence of every existing row, the encoder, the reference banks, the denominator floor, and the coefficients in Eq.* (11)*. Appending new rows leaves the score of every existing row unchanged. Hence, every pairwise score difference and the maximizing row restricted to the existing vocabulary are unchanged, using a fixed tie-breaking rule.*

Each existing raw score, availability mask and base-reference statistic stays fixed, so their weighted sum stays fixed. The proposition specifies this structural invariant; it does not by itself imply a recognition gain, and raw or fixed-temperature fusion can satisfy the same invariant. This applies to base and previously appended novel rows. Supplementary Section E provides the full proof and the consequence for expanded predictions. The implementation computes the fixed base text block separately from novel columns to keep its numerical kernel independent of the appended width.

4.5. Base-only calibration

Fusion calibration must respect the distinction between base supervision and novel support. We construct a pseudo-expansion episode by splitting the base classes into approximately equal pseudo-base and pseudo-novel groups: 20/20 on CIFAR-100, 7/8 on Pets and 50/50 on ImageNet-R. A separate classifier is fitted using the pseudo-base classes alone. The pseudo-novel rows contain names and k support examples from those base classes, while held-out base validation images supply query labels. This episode simulates missing learned evidence without consuming any actual novel-class validation labels.

For each support budget, candidate coefficients are evaluated on this episode. The selection objective is the harmonic mean of macro accuracy on the two groups,

$$H = \frac{2A_B A_N}{A_B + A_N}, \tag{14}$$

with $H = 0$ if both terms vanish. The main experiment uses $a \in \{0, 0.25, 0.5, 1, 2\}$, $u \in \{0, 0.25, 0.5, 1, 2, 4\}$, and offsets $d \in [-4, 4]$ in steps of 0.25. Ties follow a fixed grid order. The full base classifier is then combined with actual novel rows using the selected coefficients. Neither actual novel validation queries nor test labels enter this selection rule.

This procedure uses one pseudo-class split per true partition and the same support-index seed as the evaluated support condition. CIFAR and Pets pseudo splits use seed $100 + p$ for class-partition seed p; ImageNet-R selects half the base classes with seed $900 + p$. Coefficients

are chosen once for a support budget and stay fixed when the vocabulary-size stress test appends successive subsets of novel classes.

4.6. Storage and computation

The frozen implementation uses one image feature per input. Class text anchors, base classifier weights, visual references, and novel prototypes are cached. For feature dimension D, the base reference banks require $O(|\mathcal{B}|D)$ storage; each novel row requires $O(D)$ additional storage for its prototype and text anchor. Once its support images have been encoded, constructing a novel prototype requires $O(kD)$ arithmetic and no optimization state.

Scoring K candidates requires $O(KD)$ arithmetic in addition to the fixed reference calculations. Image encoding is shared across the vocabulary, and existing row scores can be reused when appending candidates for the same query.

5. Experiments

5.1. Data, access conditions, and implementation

We evaluate expansion on CIFAR-100 [28] with long-tailed base supervision. The training class counts follow an exponential schedule with imbalance parameter $\rho = 100$ and nominal maximum 500 images. Five images per class are reserved before constructing the long-tailed pool, so the largest available pool contains 495 images. The remaining long-tailed pool contains 10,894 images, split into 10,487 fitting images and 407 held-out validation images. Validation reserves up to five examples per class, with smaller allocations for the rarest classes.

For seeds $\{0, \ldots, 9\}$, stratified sampling within head, medium, and tail frequency buckets yields distinct partitions of 40 base, 40 novel, and 20 unused classes. Proportional quotas use largest-remainder rounding after shuffling class membership. Evaluation uses the 80 base and novel classes.

Oxford-IIIT Pets [29] provides a complementary fine-grained task with 37 cat and dog breeds. We use its official trainval/test split and reserve five training images per breed as support. From the remaining training images, we construct an exponential long-tailed pool with $\rho = 10$ and maximum class count 90, giving 1,168 fitting and 146 validation images. Each true partition contains 15 base, 15 novel, and seven unused breeds, sampled with the same allocation rule. The official test split contains 3,669 images. Unlike CIFAR, its images have varied spatial resolutions; RGB images are resized to 224×224 by PIL bicubic interpolation before the same CLIP normalization. These features are freshly extracted for this study.

All primary expansion readouts use frozen OpenAI CLIP ViT-L/14 or ViT-B/32 features from a single 224×224 bicubic-resized image with CLIP channel normalization. Fitting, validation, and support pools are disjoint. Supplementary Section C gives feature-cache and image-pool verification details.

Base ridge regularization is selected from $\{0.01, 0.1, 1, 10\}$ by base validation macro accuracy; the pseudo-base classifier uses the same rule. The primary comparison uses all five reserved support images, ten distinct class partitions and two arrival orders per partition. Earlier one/two/five-shot and repeated-calibration studies use three original partitions and are reported separately in Supplementary Section G.

Classifier fitting, regularization selection, and gate calibration use base classes exclusively. Five-shot CIFAR expansion consumes 200 support images for 40 novel classes. Its balanced test set supplies 4,000 base and 4,000 novel images for the expanded metrics.

ImageNet-R [30] extends the active vocabulary to 200 rendition classes: 100 base classes followed by five arrivals of 20 classes. After removing 59 byte-identical duplicates, per-class seeded splits reserve approximately 20% each for validation and test, then five support images. The fitting pool is downsampled by class rank with a nominal 100:1 schedule, yielding 2–136 images per class after availability caps. The resulting pools contain 6,438 fitting, 5,911 validation, 1,000 support and 5,911 test images. Ten frequency-stratified class partitions and two arrival orders use the same selection and reporting rules as the other datasets. RGB images are resized to 224×224 with bicubic interpolation and encoded by the same selected frozen checkpoint within each encoder comparison.

5.2. Compared readouts and statistical unit

Every readout shares its encoder, fitting pool, class partition and novel support. The retained table combines text and trained columns for base classes with text and support prototypes for novel classes. Fixed-gate removal sets the learned coefficient to zero; recalibrated removal independently reselects its remaining mixture. Dense anchored text–prototype fusion supplies visual means for every class. Tip-Adapter [8] supplies dense exponential semantic-cache retrieval from all permitted base fitting images and novel support. Both its published summed retrieval and an explicit class-mean control for unequal cache counts use the previously specified base-only selection grids. Recent FeNeC readouts [24] remain a three-partition controlled comparison in Supplementary Section F; their published-protocol numbers are not mixed with the expanded comparison.

Two simpler retained-classifier controls isolate normalization. Raw fusion uses cosine text/prototype scores and unnormalized ridge outputs. Fixed-temperature fusion divides each branch by one scalar, the mean per-query standard deviation across its fixed base reference columns on base-validation features. Scales are estimated once for the pseudo episode and once for the full base classifier, floored at 10^{-6}, and remain fixed throughout arrivals. These controls use the same coefficient/offset grids and pseudo-episode objective as the retained table. All three constructions can preserve existing scores; their recognition outcomes test the benefit of per-query reference statistics.

Two arrival orders use seeds $1000 + p$ and $1100 + p$ for partition p. Orders are averaged within each partition. We report means and sample standard deviations over ten distinct class assignments; the original three assignments are retained inside this expanded set. Paired differences use identical partition/order/session cells and, for constrained comparisons, their jointly feasible intersection. Percentile bootstrap intervals use 20,000 resamples of the paired partition means with seed 2026. These conditional intervals describe class-partition variation on shared fixed image pools and encoders, not independent-image or pretraining uncertainty. Supplementary Section I gives the full protocol and tables.

5.3. Validation-selected admission and achieved retention

For retention ratio r, each readout receives the same absolute target $\tau_r = rA^{\text{closed}}_{B,\text{ref}}$, using the retained table's closed-base accuracy on validation. We select the largest feasible group offset from the margins of correctly classified base-validation queries, then apply it to test queries without actual novel validation or test labels. A target above a readout's within-base ceiling is infeasible. We report both test group accuracies and separately count whether $A_{B,\text{test}} \geq rA^{\text{closed}}_{B,\text{ref,test}}$. That test criterion evaluates the fixed policy; it never selects its threshold.

On ImageNet-R, all ten partitions are validation-feasible for the retained, dense-prototype and class-mean cache readouts. Their novel test accuracies are 84.63%, 83.94% and 81.61%, with base test accuracies 86.15%, 86.01% and 86.08%, respectively (Table 1). The retained policy attains the separate 95% test target in 2 of ten partitions. Validation feasibility is therefore reported independently of realized test retention. The smaller-query study in Supplementary Section H measures threshold-label demand with the original fusion calibration held fixed.

Table 1: Validation-selected 95% retention policy with ViT-L/14 after five arrivals. Accuracies are percentages, mean ± sample standard deviation over feasible class partitions. Feasible counts validation-feasible partitions; Test met counts those attaining the separate test-retention criterion. All readouts share the same ten assignments and per-partition target; available means can cover different feasible subsets. The full ViT-B/32 and eight-readout comparisons appear in Supplementary Section I.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	84.26 ± 1.91	77.10 ± 2.53	10/10	7/10
CIFAR-100	Recalibrated removal	82.91 ± 1.62	66.03 ± 13.04	6/10	1/6
CIFAR-100	Dense text + prototypes	84.00 ± 1.76	75.10 ± 7.15	8/10	4/8
CIFAR-100	Tip-Adapter mean	83.19 ± 1.72	68.69 ± 10.39	9/10	2/9
CIFAR-100	Raw-score fusion	82.11 ± 2.51	64.45 ± 13.35	7/10	1/7
CIFAR-100	Fixed temperatures	83.72 ± 2.24	70.93 ± 10.56	10/10	5/10
Pets	Retained references	89.95 ± 2.67	93.79 ± 3.83	10/10	1/10
Pets	Recalibrated removal	90.48 ± 3.04	93.44 ± 3.56	8/10	3/8
Pets	Dense text + prototypes	89.46 ± 3.08	94.47 ± 2.60	10/10	2/10
Pets	Tip-Adapter mean	90.73 ± 2.17	91.39 ± 4.32	10/10	4/10
Pets	Raw-score fusion	91.35 ± 2.91	92.72 ± 2.70	9/10	4/9
Pets	Fixed temperatures	90.48 ± 3.42	94.21 ± 3.19	9/10	3/9
ImageNet-R	Retained references	86.15 ± 0.94	84.63 ± 0.95	10/10	2/10
ImageNet-R	Recalibrated removal	86.16 ± 0.71	66.88 ± 22.79	10/10	3/10
ImageNet-R	Dense text + prototypes	86.01 ± 0.94	83.94 ± 1.10	10/10	3/10
ImageNet-R	Tip-Adapter mean	86.08 ± 1.08	81.61 ± 1.51	10/10	4/10
ImageNet-R	Raw-score fusion	86.15 ± 0.68	65.55 ± 20.42	10/10	2/10
ImageNet-R	Fixed temperatures	85.99 ± 0.71	82.92 ± 1.03	10/10	1/10

5.4. Classifier value over successive arrivals

Five sessions add eight CIFAR classes, three Pets breeds or 20 ImageNet-R classes at each arrival. With the base-only selected offset fixed throughout, $\overline{A} = 5^{-1} \sum_{t=1}^{5} A_{\mathrm{all},t}$ is average incremental macro accuracy; final accuracy uses the complete active vocabulary. Table 2 reports paired retained-minus-comparator differences, separating recalibration after removal from replacement by dense prototypes or semantic caches.

Table 2: Paired classifier value with fixed base-only calibrated gates and offsets. Entries are mean gains in percentage points [conditional paired 95% bootstrap CI] across ten class partitions after averaging the two arrival orders. Positive values favor retention. The interval resamples class assignments on the fixed data; it is not a population-performance guarantee.

Encoder	Data	Comparator	Paired avg. gain [95% CI]	Paired final gain [95% CI]
L/14	CIFAR-100	Recalibrated removal	3.00 [2.23, 3.75]	2.62 [1.81, 3.43]
L/14	CIFAR-100	Dense text + prototypes	0.15 [-0.46, 0.64]	-0.21 [-0.85, 0.30]
L/14	CIFAR-100	Tip-Adapter mean	1.44 [0.68, 2.10]	1.24 [0.41, 2.01]
L/14	Pets	Recalibrated removal	-0.14 [-1.53, 0.86]	-0.78 [-2.84, 0.60]
L/14	Pets	Dense text + prototypes	-0.98 [-2.52, 0.22]	-1.54 [-4.08, 0.35]
L/14	Pets	Tip-Adapter mean	-0.06 [-1.76, 1.41]	-0.39 [-2.92, 1.71]
L/14	ImageNet-R	Recalibrated removal	2.30 [1.95, 2.59]	1.71 [1.28, 2.06]
L/14	ImageNet-R	Dense text + prototypes	0.71 [0.46, 0.92]	0.41 [0.03, 0.72]
L/14	ImageNet-R	Tip-Adapter mean	1.42 [1.15, 1.62]	1.13 [0.76, 1.43]
B/32	CIFAR-100	Recalibrated removal	2.38 [1.44, 3.34]	1.80 [1.01, 2.65]
B/32	CIFAR-100	Dense text + prototypes	0.86 [-0.15, 1.76]	0.43 [-0.47, 1.24]
B/32	CIFAR-100	Tip-Adapter mean	0.96 [-0.04, 1.84]	0.51 [-0.34, 1.33]
B/32	Pets	Recalibrated removal	-0.44 [-1.64, 0.81]	-2.00 [-3.80, -0.19]
B/32	Pets	Dense text + prototypes	-0.39 [-1.85, 1.02]	-1.17 [-3.22, 0.67]
B/32	Pets	Tip-Adapter mean	-0.74 [-1.62, 0.12]	-1.85 [-3.40, -0.59]
B/32	ImageNet-R	Recalibrated removal	3.07 [2.82, 3.34]	2.39 [2.19, 2.61]
B/32	ImageNet-R	Dense text + prototypes	0.90 [0.69, 1.11]	0.46 [0.26, 0.66]
B/32	ImageNet-R	Tip-Adapter mean	1.71 [1.44, 1.98]	1.34 [1.16, 1.52]

On ImageNet-R, the ViT-L/14 retained table reaches 87.24% average incremental accuracy. Its paired gains are 0.71 [0.46, 0.92] points over dense prototypes and 1.42 [1.15, 1.62] over class-mean cache retrieval. With ViT-B/32, these gains are 0.90 [0.69, 1.11] and 1.71 [1.44, 1.98] points. The ImageNet-R gain therefore transfers across the two encoder architectures under identical image-pool access. CIFAR gains over classifier-free recalibration also appear with both encoders, while the dense-prototype comparison has intervals spanning zero. On

Pets, average incremental intervals span zero for all three comparators with both encoders; the final B/32 comparisons can favor the alternatives. These regimes distinguish the value of stored learned evidence from a claim of universal accuracy superiority.

5.5. Matched-test-retention capacity and simple fusion

The descriptive frontier fixes each mixture and asks

$$N_r = \max_d A_N(d) \quad \text{subject to} \quad A_B(d) \geq r A_{B,\text{ref,test}}^{\text{closed}}. \tag{15}$$

This sweep uses base test labels to characterize attainable capacity; it is separate from the validation-selected policy above. Sorting correctly classified base-query margins with macro weights identifies the largest feasible offset; Supplementary Section D derives its exact empirical envelope. We evaluate $r \in \{0.80, 0.85, 0.90, 0.95, 0.99\}$, with the same per-partition reference target for every readout. Infeasible targets are missing with explicit counts.

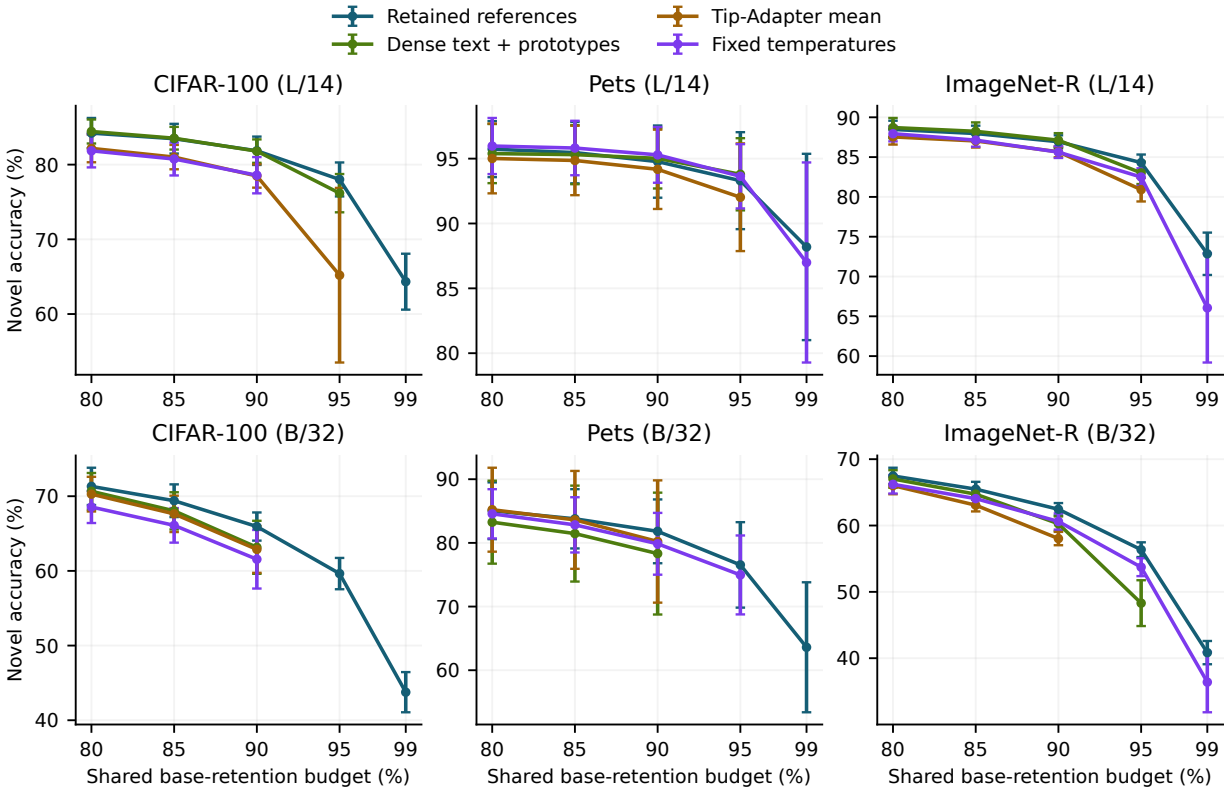


Figure 2: Descriptive novel-recognition frontiers over ten class partitions with two frozen CLIP encoders. Every readout receives the retained model's common absolute target within a partition and encoder. Error bars are partition-level sample standard deviations after averaging arrival orders. A point is drawn only when all ten partitions are feasible; Supplementary Section I reports available partial sets and counts. Thresholds use test labels for this capacity analysis and are not deployment calibration.

Table 3: Normalization controls with ViT-L/14. Fixed-gate average covers five arrivals; matched base/novel accuracy uses the descriptive 95% test-retention budget after the fifth arrival. Values are percentages, mean ± partition-level sample standard deviation; counts identify feasible sets. All fusion controls share the pseudo-episode grids and retained base classifier.

Data	Fusion	Fixed-gate avg.	Matched base	Matched novel	Feasible
CIFAR-100	Retained references	83.03 ± 1.19	83.91 ± 1.42	78.02 ± 2.27	10/10
CIFAR-100	Raw-score fusion	79.70 ± 2.64	83.25 ± 1.17	62.37 ± 6.38	6/10
CIFAR-100	Fixed temperatures	81.56 ± 0.98	83.64 ± 1.21	74.53 ± 2.70	9/10
Pets	Retained references	92.23 ± 2.87	91.56 ± 1.87	93.30 ± 3.74	10/10
Pets	Raw-score fusion	87.94 ± 14.58	91.57 ± 1.88	92.39 ± 2.75	10/10
Pets	Fixed temperatures	89.39 ± 10.15	91.56 ± 1.87	93.64 ± 2.48	10/10
ImageNet-R	Retained references	87.24 ± 0.59	86.42 ± 0.53	84.31 ± 1.01	10/10
ImageNet-R	Raw-score fusion	83.70 ± 2.58	86.42 ± 0.56	70.85 ± 8.03	9/10
ImageNet-R	Fixed temperatures	86.30 ± 0.38	86.43 ± 0.53	82.49 ± 1.11	10/10

At the ImageNet-R 95% budget, the retained ViT-L/14 readout reaches 86.42% base and 84.31% novel accuracy. Relative to fixed-temperature fusion, its paired average incremental gain is 0.94 [0.74, 1.12] points. This comparison isolates per-query branch scaling from the append-only invariant shared by both constructions. The complete budget curves and simple controls show where recognition value accompanies score preservation, while the Pets comparison separates structural preservation from an accuracy advantage. The imbalance schedules are held fixed in these experiments; differences across datasets and encoders describe the observed regimes rather than a causal effect of changing the imbalance ratio.

6. Discussion

Retained discrimination has an admission value beyond its closed-base accuracy. Under a shared base requirement, a classifier that separates the original classes more effectively can admit novel candidates at a more permissive threshold. The matched frontiers express this value directly as novel accuracy, while their feasible regions identify the base service levels that each readout can support.

Sequential comparisons make the contribution measurable across repeated arrivals. Fixed-gate removal evaluates the learned branch inside a selected mixture; classifier-free recalibration measures the value that remains after restoring the mixture's operating point; dense base

prototypes provide a stronger replacement alternative. These distinctions connect retained parameters to recognition outcomes rather than treating parameter preservation as the final objective.

Reference anchoring supplies a consistent score frame for this evaluation. Stored base columns and reference statistics remain fixed, branch coefficients transfer from base-only episodes, and the group threshold controls novel admission. A fixed threshold gives append-only score preservation; validation-based threshold adjustment supplies an explicit retention policy. Both operate on the same stored classifier.

7. Conclusion

We presented retention-constrained class expansion with reference-anchored candidate tables. The formulation measures novel recognition at a common base-accuracy requirement and quantifies the retained classifier's paired contribution across successive arrivals. Retention frontiers quantify the operating regimes in which learned base discrimination supports novel admission; five-session comparisons separate this contribution from calibration and prototype replacement. Preserving the original classifier becomes a measurable source of recognition capacity within a consistent expandable score system.

CRediT authorship contribution statement

Junyi Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The experiments use public CIFAR-100, Oxford-IIIT Pets and ImageNet-R data and the publicly released CLIP ViT-L/14 checkpoint. Code, source-hashed experimental records, prediction archives, manuscript sources, and table/figure generation scripts are available at `https://github.com/Cheerssom/Reference-Anchored-Candidate-Tables`. Dataset images, pretrained weights, and feature caches are obtained separately using the repository instructions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with literature discovery, manuscript drafting and revision, experiment and verification scripts, and LaTeX and table preparation. The author is responsible for the accuracy, interpretation and originality of the submitted work and for reviewing and approving the final manuscript and supporting materials.

References

[1] Z. Liu, Z. Miao, X. Zhan, J. Wang, B. Gong, S. X. Yu, Large-Scale Long-Tailed Recognition in an Open World, arXiv preprint arXiv:1904.05160 (2019).
URL `https://arxiv.org/abs/1904.05160`

[2] K. Cao, C. Wei, A. Gaidon, N. Arechiga, T. Ma, Learning Imbalanced Datasets with Label-Distribution-Aware Margin Loss, arXiv preprint arXiv:1906.07413 (2019).
URL `https://arxiv.org/abs/1906.07413`

[3] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, S. Belongie, Class-Balanced Loss Based on Effective Number of Samples, arXiv preprint arXiv:1901.05555 (2019).
URL `https://arxiv.org/abs/1901.05555`

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning Transferable Visual Models From Natural Language Supervision, in: International Conference on Machine Learning, 2021, pp. 8748–8763.
URL `https://proceedings.mlr.press/v139/radford21a.html`

[5] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Learning to Prompt for Vision-Language Models, International Journal of Computer Vision 130 (9) (2022) 2337–2348. doi:10.1007/s11263-022-01653-1.
URL https://link.springer.com/article/10.1007/s11263-022-01653-1

[6] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Conditional Prompt Learning for Vision-Language Models, arXiv preprint arXiv:2203.05557 (2022).
URL https://arxiv.org/abs/2203.05557

[7] J.-X. Shi, T. Wei, Z. Zhou, J.-J. Shao, X.-Y. Han, Y.-F. Li, Long-Tail Learning with Foundation Model: Heavy Fine-Tuning Hurts, in: International Conference on Machine Learning, 2024, pp. 45014–45039.
URL https://proceedings.mlr.press/v235/shi24g.html

[8] R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, H. Li, Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification, arXiv preprint arXiv:2207.09519 (2022).
URL https://arxiv.org/abs/2207.09519

[9] J. Snell, K. Swersky, R. S. Zemel, Prototypical Networks for Few-shot Learning, arXiv preprint arXiv:1703.05175 (2017).
URL https://arxiv.org/abs/1703.05175

[10] Y. Wang, W.-L. Chao, K. Q. Weinberger, L. van der Maaten, SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning, arXiv preprint arXiv:1911.04623 (2019).
URL https://arxiv.org/abs/1911.04623

[11] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, C. H. Lampert, iCaRL: Incremental Classifier and Representation Learning, arXiv preprint arXiv:1611.07725 (2016).
URL https://arxiv.org/abs/1611.07725

[12] X. Tao, X. Hong, X. Chang, S. Dong, X. Wei, Y. Gong, Few-Shot Class-Incremental Learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 12183–12192.
URL https://openaccess.thecvf.com/content_CVPR_2020/html/Tao_Few-Shot_Class-Incremental_Learning_CVPR_2020_paper.html

[13] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, D.-C. Zhan, Z. Liu, Revisiting Class-Incremental Learning with Pre-Trained Models: Generalizability and Adaptivity are All You Need, International Journal of Computer Vision (2024).
URL https://github.com/LAMDA-CL/RevisitingCIL

[14] M. D. McDonnell, D. Gong, A. Parvaneh, E. Abbasnejad, A. van den Hengel, RanPAC: Random Projections and Pre-trained Models for Continual Learning, in: Advances in Neural Information Processing Systems, Vol. 36, 2023, pp. 12022–12053. doi:10.52202/075280-0526.
URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/2793dc35e14003dd367684d93d236847-Abstract-Conference.html

[15] X. Xu, S. Niu, Z. Wang, W. Guo, L. Jing, H. Yang, Multi-feature space similarity supplement for few-shot class incremental learning, Knowledge-Based Systems 265 (2023) 110394. doi:10.1016/j.knosys.2023.110394.

[16] R. Chen, X.-Y. Jing, F. Wu, H. Chen, Sharpness-aware gradient guidance for few-shot class-incremental learning, Knowledge-Based Systems 299 (2024) 112030. doi:10.1016/j.knosys.2024.112030.

[17] H. Fang, H. Cai, K. Mao, J. Zhao, X. Xiao, Few-shot class-incremental learning based on prompt guidance and multimodal fusion, Knowledge-Based Systems 342 (2026) 115862. doi:10.1016/j.knosys.2026.115862.

[18] C. Han, Z. Gu, C. Xu, Z. Cui, Adaptive exploration for few-shot incremental learning, Knowledge-Based Systems 330 (2025) 114496. doi:10.1016/j.knosys.2025.114496.

[19] Q. Wang, K. Xu, D. Feng, B. Ding, H. Wang, Memory replay with unlabeled data for semi-supervised class-incremental learning via temporal consistency, Frontiers of Computer Science 19 (12) (2025) 1912370. doi:10.1007/s11704-025-40828-0.
URL https://journal.hep.com.cn/fcs/EN/10.1007/s11704-025-40828-0

[20] J. Tang, X. Wang, S. Chen, Global pre-fixing, local adjusting: a simple yet effective contrastive strategy for continual learning, Frontiers of Computer Science 21 (2) (2027) 2102324. doi:10.1007/s11704-025-50623-6.
URL https://journal.hep.com.cn/fcs/EN/10.1007/s11704-025-50623-6

[21] W.-L. Chao, S. Changpinyo, B. Gong, F. Sha, An Empirical Study and Analysis of Generalized Zero-Shot Learning for Object Recognition in the Wild, arXiv preprint arXiv:1605.04253 (2016).
URL https://arxiv.org/abs/1605.04253

[22] Q. Yan, Y. Yang, Y. Dai, X. Zhang, K. Wiltos, M. Woźniak, W. Dong, Y. Zhang, CLIP-guided continual novel class discovery, Knowledge-Based Systems 310 (2025) 112920. doi:10.1016/j.knosys.2024.112920.

[23] R. He, D. Fang, Y. Chen, K. Tong, C. Chen, Y. Wang, L.-p. Chau, H. Zhuang, REAL: Representation enhanced analytic learning for exemplar-free class-incremental learning, Knowledge-Based Systems 332 (2026) 114901. doi:10.1016/j.knosys.2025.114901.

[24] K. Książek, H. Jastrzębski, K. Pniaczek, B. Trojan, M. Karp, J. Tabor, FeNeC: Enhancing continual learning via feature clustering with neighbor- or logit-based classification, Knowledge-Based Systems 338 (2026) 115479. doi:10.1016/j.knosys.2026.115479.

[25] Y. Chen, T. Ding, L. Wang, J. Huo, Y. Gao, W. Li, Enhancing few-shot class-incremental learning via training-free bi-level modality calibration, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 9881–9890.
URL https://openaccess.thecvf.com/content/CVPR2025/html/Chen_Enhancing_Few-Shot_Class-Incremental_Learning_via_Training-Free_Bi-Level_Modality_Calibration_CVPR_2025_paper.html

[26] E. Lee, M. Kim, J. Kwon, Y. Lee, J. Kim, S. Jang, Y. Kim, Hycal: A training-free prototype calibration method for cross-discipline few-shot class-incremental learning, arXiv:2604.15678v2 (2026).
URL https://arxiv.org/abs/2604.15678

[27] D.-W. Zhou, Z.-H. Qi, H.-J. Ye, D.-C. Zhan, TV100: a TV series dataset that pre-trained CLIP has not seen, Frontiers of Computer Science 18 (5) (2024) 185349. doi:10.1007/s11704-024-40217-z.
URL https://journal.hep.com.cn/fcs/EN/10.1007/s11704-024-40217-z

[28] A. Krizhevsky, Learning multiple layers of features from tiny images, Tech. rep., University of Toronto (2009).
URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf

[29] O. M. Parkhi, A. Vedaldi, A. Zisserman, C. V. Jawahar, Cats and dogs, in: IEEE Conference on Computer Vision and Pattern Recognition, 2012, pp. 3498–3505. doi:10.1109/CVPR.2012.6248092.
URL https://www.robots.ox.ac.uk/~vgg/publications/2012/parkhi12a/

[30] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, J. Gilmer, The Many Faces of Robustness: A Critical Analysis of Out-of-Distribution Generalization, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 8340–8349.
URL https://openaccess.thecvf.com/content/ICCV2021/html/Hendrycks_The_Many_Faces_of_Robustness_A_Critical_Analysis_of_Out-of-Distribution_ICCV_2021_paper.html